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B Tech CSE {Core}
DBMS Assignment - 1

Sec - A → Introduction to DBMS.

Q1. What is Database Management System? Why it is required?

Ans. A Database Management System (DBMS) is a software system used to store, retrive and run queries on data. It acts as an Interface between the user and the database.
Managing how data is stored and Accessed:
It is required for several reasons:
(i) Data Management
(ii) Data Sharing
(iii) Security
(iv) Backup and Recovery

Q2. Why is DBMS better than traditional file system?

Ans. DBMS offers significant advantages over Traditional file system by Addressing their inpresent limitations.
Key reasons include:

(i) Data redundancy and inconsistency: DBMS reduces data duplication and Ensures data consistency across the system.

(ii) Data sharing: DBMS facilitates easy data sharing among users and Manages simultaneous access effectively.

(iii) Data Abstraction: It offers users a simplified view of the data, hiding complex storage details.

(iv) Backup and Recovery: DBMS includes built-in feature for data backup and recovery.

Q3. What is Data Abstraction? Explain its level.

Ans. Data abstraction is the process of hiding complex data storage and retrieval details from the users, providing a simplified Abstract view of the data.

There are three level of Data Abstraction:

(i) Physical level: This is the lowest level, describing how the data is actually stored in the physical storage media.

(ii) Conceptual level: This level describes what data is stored in the database and the relationship among the data.

(iii) View level: This is the highest level, describing only a part of database that a particular usergroup needs.

Q4. What is three schema Architecture?

Ans. The Three schema Architecture is a framework for database design that separates the user Applications from the physical database, utilizing the three level of data Abstraction: External, Conceptual & Internal.

Importance:

(i) Data Independence

(ii) Multiple User Views

(iii) Data Security

(iv) Standardization.

Q5. What is Data Independence?

Ans. Data Independence is the ability to modify the schema definition at one level of the DBMS without affecting the schema definition at the Next higher level.

It is Needed for:

(i) Flexibility: It allows database Adminstrator to modify the physical storage structures or conceptual schema: without having to rewrite application programs.

(ii) Maintenance: It simplifies system Maintainence & Evaluation over time.

Q6. What is a ER Model?

Ans. An Entity Relationship {ER} model is a high level conceptual data model used to represent the structure of a database graphically.

It is used for:

(i) Design Database: It helps in Designing the database by identifying the Entities their Attributes and the relationship between them.

(ii) Visualise Data: It provides a clear visual representation of the data structure, making it Easier to understand and communicate.

Q7. What are Integrity Constraints?

Ans. Integrity Constraints are rules that restrict that data values that can be entered into database, Ensuring data Accuracy and Consistency.

They are Important because:

(i) Maintain Data Quality: They prevent the Entry of invalid or Inconsistent data into the database.

(ii) Ensure Reliability: They help maintain the reliability of the information stored in the database.

Sec - B → Design, Draw and Defend.

Q1. Three - Schema Architecture Diagram:

Ans. The three-schema architecture is a proposed approach for database system, design, consisting of three levels:

(i) Internal level: This is the physical storage level. It describes how the data is actually stored in the database including File structures.

(ii) Conceptual level: This is the logical level. It describes the Entire database structure for a community of users, hiding the physical storage details.

(iii) External level: This level describes the part of the database that a particular user group is interested in.

It presents a simplified view of the database.

Q2. Online Shopping:

Ans: Primary keys: Custome ID (for the customer Table)
Product ID (For the product Table)
Order ID (For the Order Table)

Foreign Keys: These link relation tables. The Order Table would contain, Customer ID and Product ID as foreign keys.

Constraints: (i) Primary key Constraint
(ii) Foreign key Constraint
(iii) Not Null constraint
(iv) Unique Constraint

Q3. University Data Abstraction

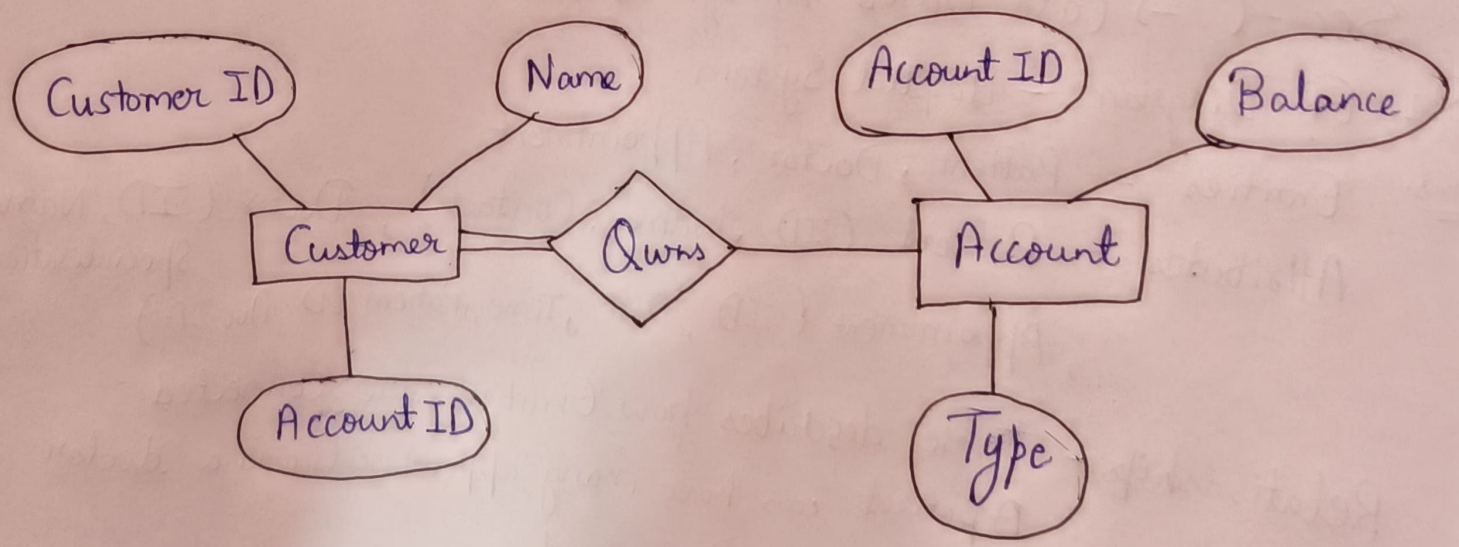
Ans. Data Abstraction helps different users by providing simplified views tailored to their needs, hiding complex details.

Here, Student see Marks {View level}
Admin Manages Records {Logical level}
DBA Control storage {Physical level}

Sec-D → One Diagram + One Insight

Q1. ER Diagram : ATM System.

Ans.



Q2. ER - diagram → Library

Ans. Basic ER diagram for a library management system includes:

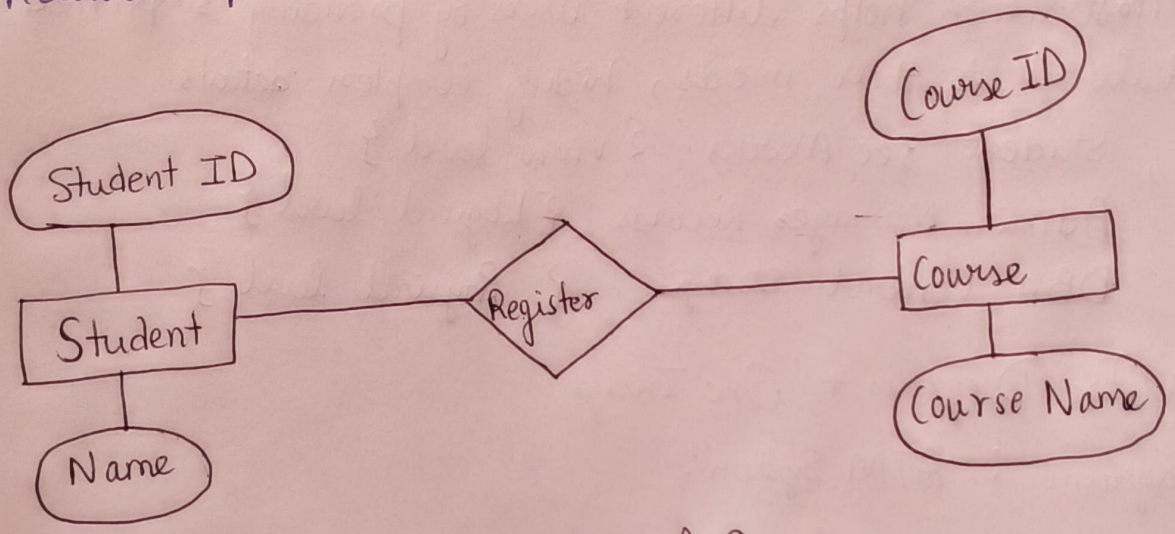
→ Entities : Member (or student), book, publishers, librarian

→ Attributes : Each Entity has attributes { Eg. Member has ID, Name : Book has ISBN, Title }

→ Relationship : ◦ A member borrows books
◦ A Publisher publishes books.
◦ A librarian manages members & books.

Q3. Student Course Registration ER.

Ans. → Student { Student ID, Name }
→ Course { Course ID, Course Name }
Relationship : Student Register For course.



Sec - C → Case Based Analytical Questions:

Q1. ER Diagram - Hospital System.

Ans. Entities - Patient, Doctor, Appointment

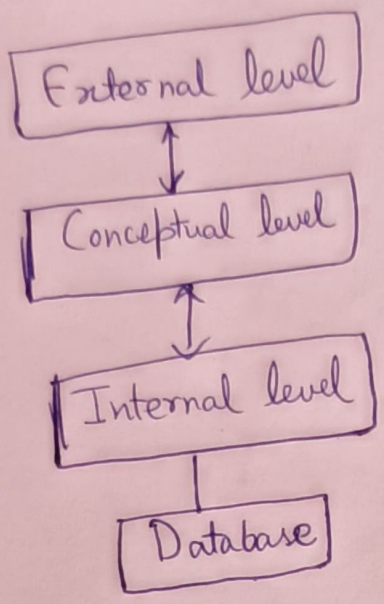
Attributes - Patient (ID, Name, Contact), Doctor (ID, Name, Specialisation), Appointment (ID, Date, Time, Patient ID, Doc ID)

Relationships - These describes how Entities are Connected.
A patient can have many appointments and a doctor can have many appointments.
This is a many to many relationship.

→ Account No. is a primary key.
→ Relation : Customer Owns Account
→ Account No. takes Uniquely Identified Account

Q2. Three Schema Architecture + One Benefit

Ans. Diagram:



Key Benefit: Data Independence → Provides high data independence meaning changes to the database structure at One level.

Q3. Draw a Entity with Attribute.

Ans. Eg: The Employee Entity
↳ Taking an Employee Entity with Employee ID as primary key.

→ Entity Employee : Employee ID
Name
Salary
Department

→ Primary Key : Here; Employee ID is referred as (PK) because for all employee their ID will be different.